



Introduction to Biostatistics

MSBD520

Fall 2026

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Housekeeping

- Office hours

Wednesdays 11 am - noon CT

Fridays 3 - 4 pm CT

- Homework 1: posted on Blackboard. Due next week (September 3).
- We will have in-class work today, make sure to upload your work by midnight.
- Starting next week, any assignments will incur a penalty for late submissions.



What did we learn last week?

Data matrix

Types of
variables

Relationship
between
variables

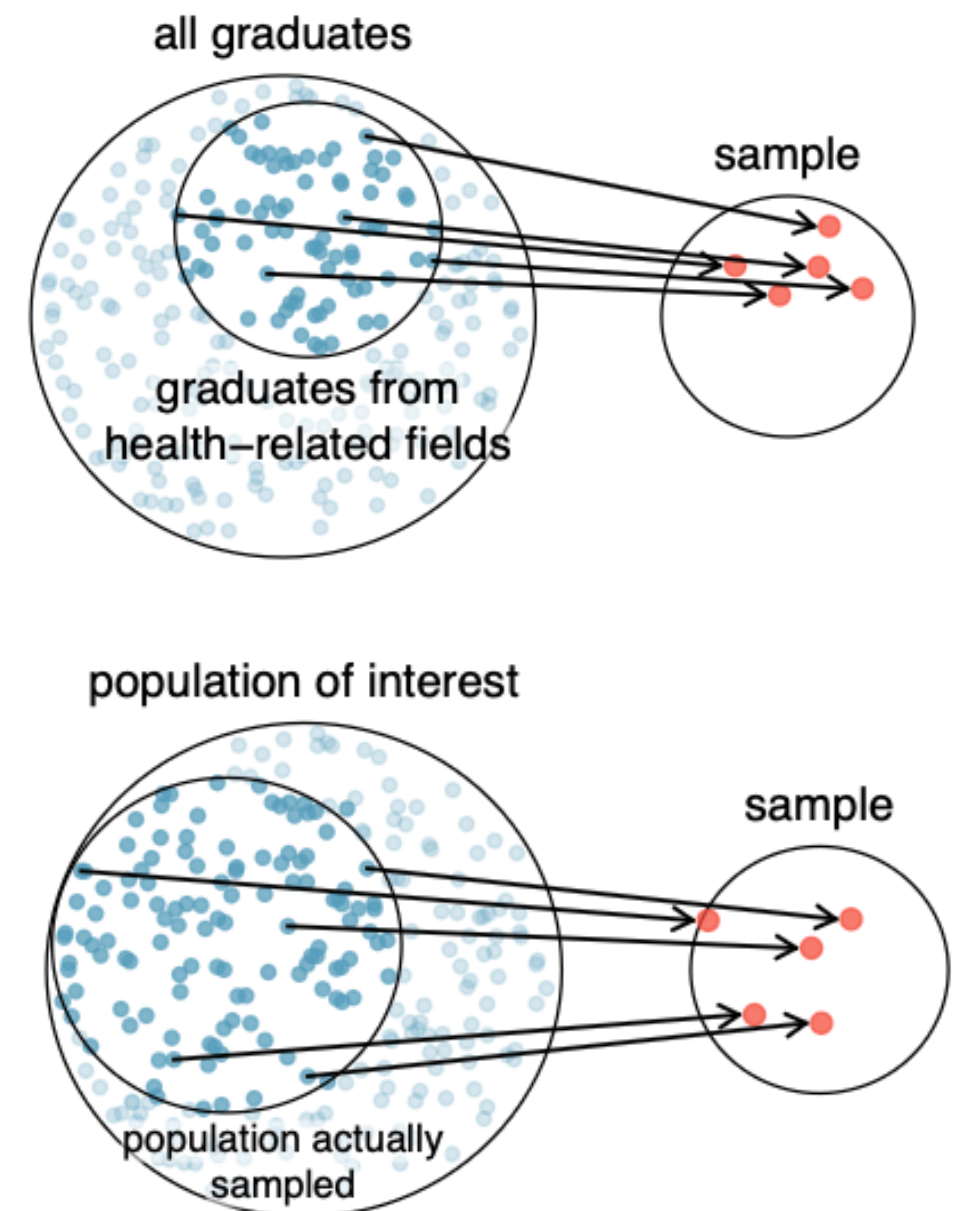
Data
collection

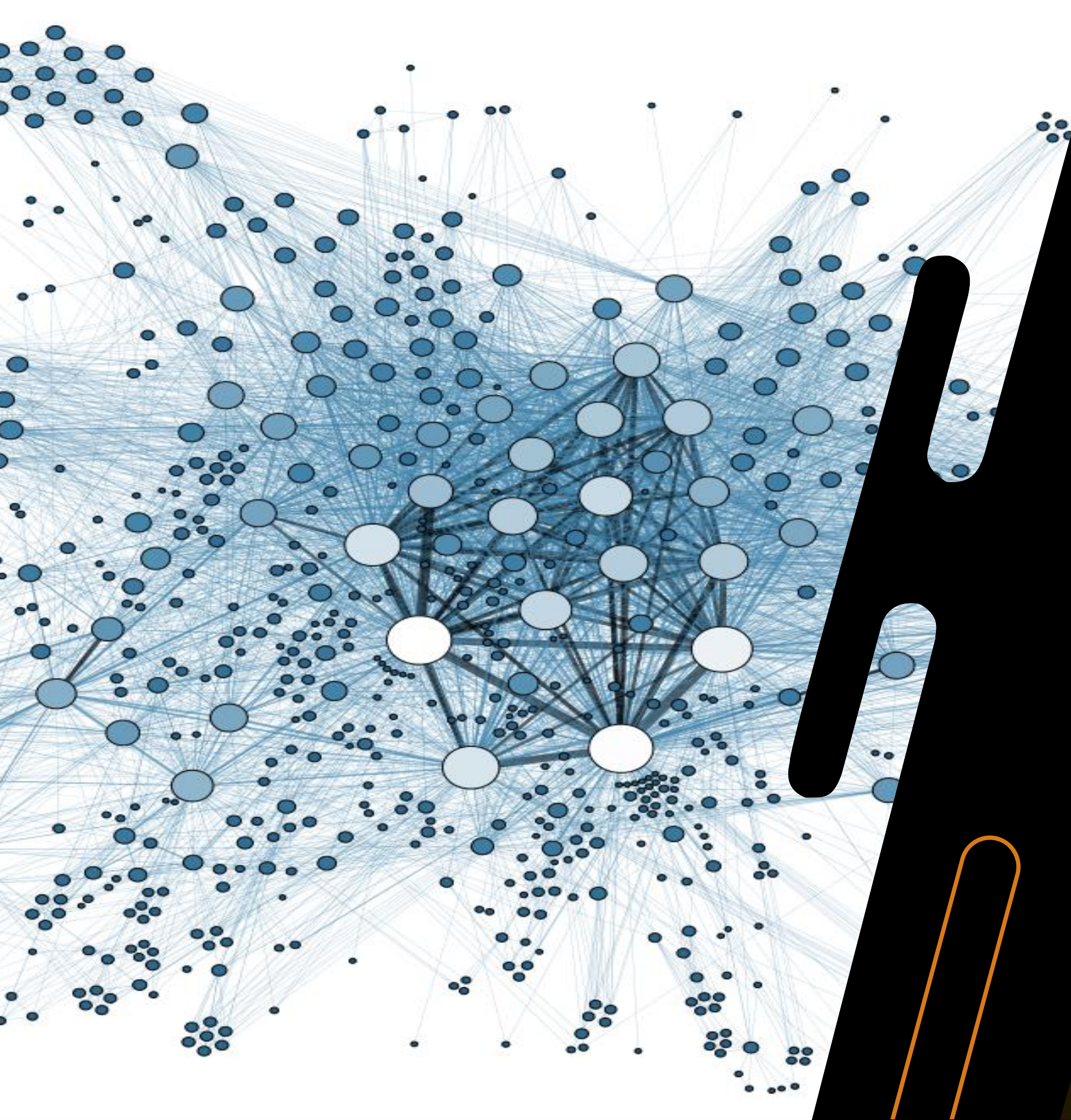
Sampling
(and sampling
bias)

Additional Points on Sampling...

Issues when we do sampling

- ❑ **Sampling Bias:** pick and choose can cause skewed population.
- ❑ **Non-response bias:** If only small fraction of the randomly sampled people choose to respond to a survey, the sample may no longer be representative of the population.
- ❑ **Convenience sampling:** individuals who are easily accessible are more likely to be included in the sample.





Experiments

Principles of experimental design



1. Control: Control for the (potential) effect of variables other than the ones directly being studied.



2. Randomize: Randomly assign subjects to treatments, and randomly sample from the population whenever possible.



3. Replicate: Within a study, replicate by collecting a sufficiently large sample. Or replicate the entire study.



4. Block: If there are variables that are known or suspected to affect the response variable, first group subjects into blocks based on these variables, and then randomize cases within each block to treatment groups.

More on blocks...

Let's look at an example:

We would like to design an experiment to investigate whether energy gels make you run faster.

We set two groups of samples -

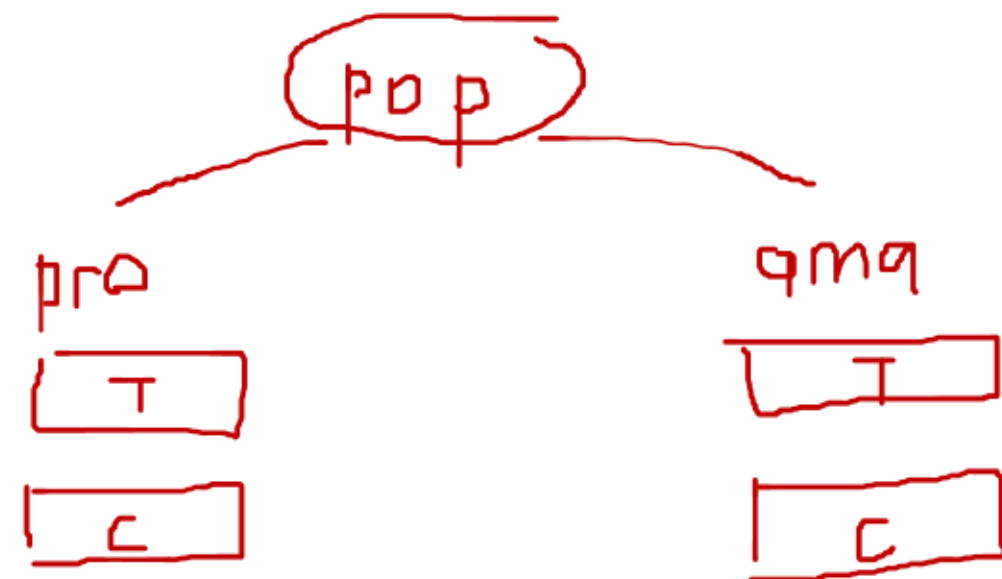
- Treatment: with energy gel
- Control: no energy gel



More on blocks...

It is suspected that energy gels might affect pro and amateur athletes differently, therefore we block for pro status:

- Divide the sample to pro and amateur
- Randomly assign pro athletes to treatment and control groups
- Randomly assign amateur athletes to treatment and control groups
- Pro/amateur status is equally represented in the resulting treatment and control groups



More on blocks...

There are differences between blocking and explanatory variables:

- Factors are conditions we can impose on the experimental units.
- Blocking variables are characteristics that the experimental units come with, that we would like to control for.
- Blocking is like stratifying, except used in experimental settings when randomly assigning, as opposed to when sampling.

Experiments

Let's try this problem: A study is designed to test the ^{exp}effect of light level and noise level ^{resp}on exam performance of students. The researcher also believes that light and noise levels ^{block}might have different effects on males and females, so wants to make sure both genders are equally represented in each group. Which of the below is correct?

- (a) There are 3 explanatory variables (light, noise, gender) and 1 response variable (exam performance)
- ☒ (b) There are 2 explanatory variables (light and noise), 1 blocking variable (gender), and 1 response variable (exam performance)
- (c) There is 1 explanatory variable (gender) and 3 response variables (light, noise, exam performance)
- (d) There are 2 blocking variables (light and noise), 1 explanatory variable (gender), and 1 response variable (exam performance)

Experiments

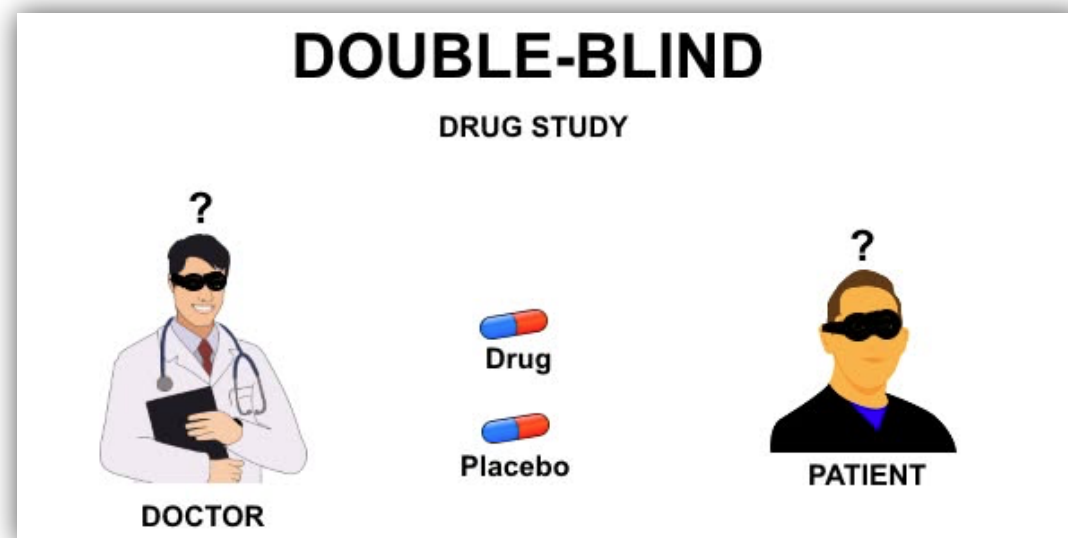
More terms in experimental design:

- **Placebo:** fake treatment, often used as the control group for medical studies
- **Placebo effect:** experimental units showing improvement simply because they believe they are receiving a special treatment
- **Blinding:** when experimental units do not know whether they are in the control or treatment group
- **Double-blind:** when both the experimental units and the researchers who interact with the patients do not know who is in the control and who is in the treatment group

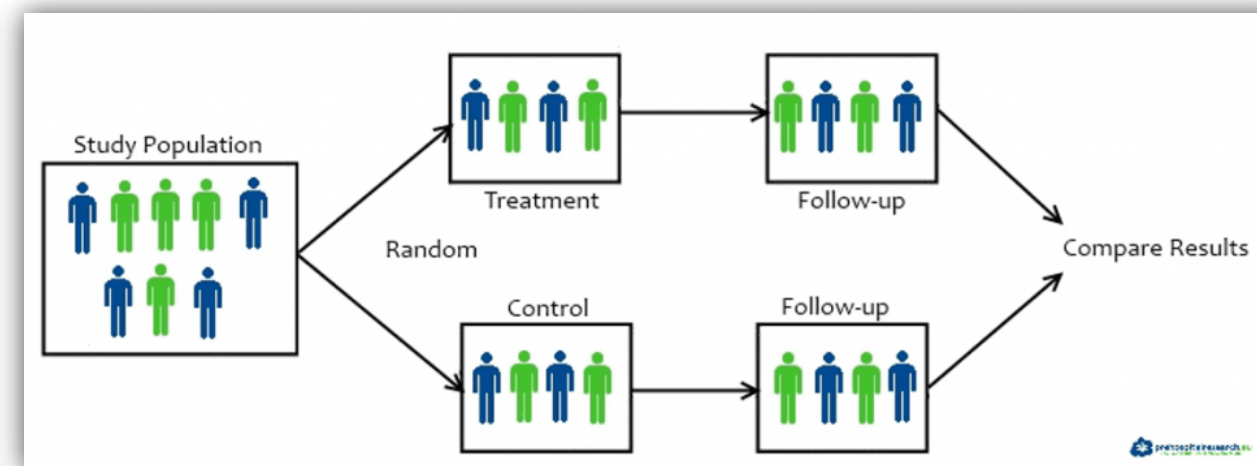
Experiments

What are some ways that can help to reduce bias in our experiment?

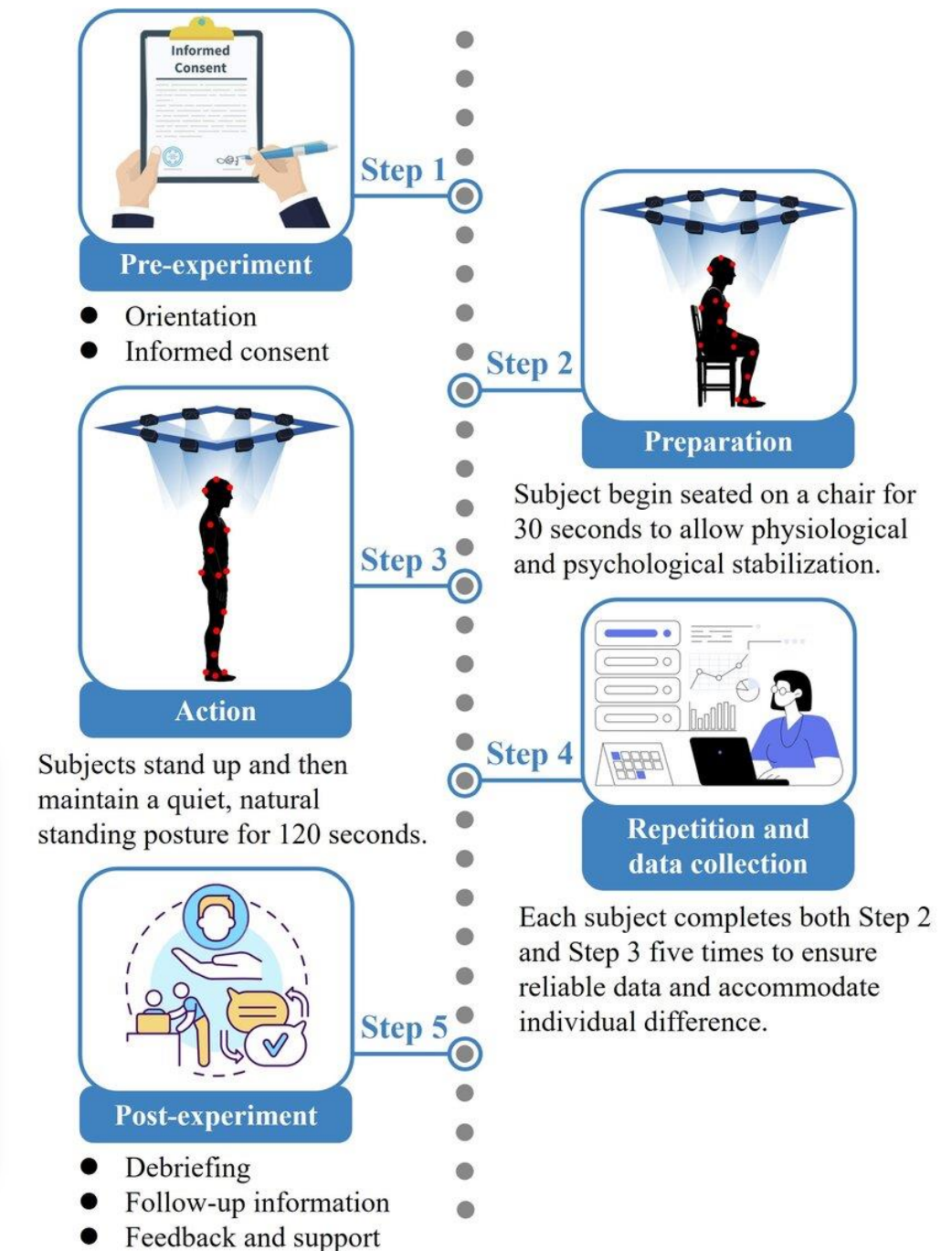
- Randomization
- Blinding
- Standardized protocol



<https://www.facebook.com/AMPWDA/photos/ampwda-double-blind-certificationfor-mantrailingwhat-is-a-double-blind-trailques/1325589587479068/>



<https://fiveable.me/college-intro-stats/unit-1/4-experimental-design-ethics/study-guide/5fOMFbJ5tujb0Onl>



https://www.researchgate.net/figure/Experimental-Protocol-The-protocol-includes-five-steps-1-Pre-experiment-orientation_fig4_385970401

Experiments

Practice:

What is the main difference between observational studies and experiments?

(a) Experiments take place in a lab while observational studies do not need to.

(b) In an observational study we only look at what happened in the past.

☒ (c) Most experiments use random assignment while observational studies do not.

(d) Observational studies are completely useless since no causal inference can be made based on their findings.

Experiments

<i>ideal experiment</i>	Random assignment	No random assignment	<i>most observational studies</i>
Random sampling	Causal conclusion, generalized to the whole population.	No causal conclusion, correlation statement generalized to the whole population.	Generalizability
No random sampling	Causal conclusion, only for the sample.	No causal conclusion, correlation statement only for the sample.	No generalizability
<i>most experiments</i>	Causation	Correlation	<i>bad observational studies</i>

Let's practice...

1.37 Chia seeds and weight loss. Chia Pets – those terra-cotta figurines that sprout fuzzy green hair – made the chia plant a household name. But chia has gained an entirely new reputation as a diet supplement. In one 2009 study, a team of researchers recruited 38 men and divided them randomly into two groups: treatment or control. They also recruited 38 women, and they randomly placed half of these participants into the treatment group and the other half into the control group. One group was given 25 grams of chia seeds twice a day, and the other was given a placebo. The subjects volunteered to be a part of the study. After 12 weeks, the scientists found no significant difference between the groups in appetite or weight loss.³³

- (a) What type of study is this?
- (b) What are the experimental and control treatments in this study?
- (c) Has blocking been used in this study? If so, what is the blocking variable?
- (d) Has blinding been used in this study?
- (e) Comment on whether or not we can make a causal statement, and indicate whether or not we can generalize the conclusion to the population at large.

- a) Experiment
- b) Treatment: 25 grams chia seeds twice a day, Control: placebo
- c) Yes, block variable is gender.
- d) Seems like it, because there is no mentioning if the volunteers are informed whether they are placebo or not.
- e) Yes, we can make causal statement because it is an experiment. But we cannot generalize to the population at large.

Case Studies: Group work

Go to Blackboard to download the case for your group.

Discuss with your peers – 15 minutes

We will get back and present our results.



Source: borisz / Getty Images